INTERNATIONAL MONETARY FUND

NEPAL

April 29, 2020

REQUEST FOR DISBURSEMENT UNDER THE RAPID CREDIT FACILITY—DEBT SUSTAINABILITY ANALYSIS

Approved By
Anne-Marie Gulde Wolfe and Maria Gonzalez (IMF) and Marcello Estevão (IDA)

Prepared by the staff of the International Monetary Fund and the International Development Association.

Nepal: Joint Bank-Fund Debt Sustainability Analysis[1]	
Risk of external debt distress	Low[2]
Overall risk of debt distress	Low
Granularity in the risk rating	Not applicable
Application of judgement	No
Macroeconomic projections	This DSA update reflects the COVID-19 shock, with a significant impact on growth in FY2019/20 and FY2020/21.[3] Remittances have fallen considerably, tourist arrivals collapsed, and domestic activities have dropped due to containment measures that include a nationwide lockdown.
Financing strategy	Given limited domestic financing capacity, Nepal seeks external financing sources to fill its financing gap. In addition to the RCF (US$214 million), the World Bank has approved a $29 million COVID-19 Emergency Response and Health Systems Preparedness Project and the Asian Development Bank is processing a pandemic response loan under the Countercyclical Support Facility for up to US$250 million. The IMF also provided debt relief under the Catastrophe Containment and Relief Trust (US$ 3.9 million). The remaining financing gap will be financed by support from development partners or reserve drawdowns. The authorities also intend to request debt service reprofiling from official bilateral creditors (including export-import banks) over the period May 1 and December 31, 2020, which would free-up resources of up to US$31 million for the authorities' COVID-19 response.
Realism tools flagged	Yes, standard multiplier analysis (that does not consider COVID-19 impact) suggests pessimistic growth in 2020 despite large fiscal stimulus.
Mechanical risk rating under the external DSA	Low
Mechanical risk rating under the public DSA	Low

[1] Debt coverage has not changed from the February 2020 DSA.

[2] Nepal's composite indicator score is 3.28, which signals a strong debt-carrying capacity.

[3] The macroframework underlying this DSA update is the same as that included in the staff report of the 2020 RCF request, based on available information at the time. However, the COVID-19 economic impact and policy response are rapidly evolving, and risks are heavily tilted to the downside. This DSA does not incorporate the potential debt service reprofiling of official bilateral debt.

This DSA updates the joint World Bank-Fund analysis of February 2020 to reflect Nepal's recent economic development, which have been considerably affected by the COVID-19 pandemic and related containment policies. Even under the COVID-19 pandemic, Nepal's debt is assessed to remain at a low risk of external and overall debt distress, unchanged from the 2020 DSA[4]. None of the debt and debt service indicators breach the indicative threshold values. However, given elevated gross financing needs in Nepal, a deeper global slowdown could negatively affect Nepal's risk of debt distress. The Nepalese authorities should continue to: (i) improve domestic productivity; (ii) streamline bureaucratic processes; (iii) compile PPP projects and private sector external debt; (iv) implement a medium-term debt strategy and develop the government bond market; and (v) improve subnational governments' public financial management and reporting, and implement a prudent framework for subnational borrowing.

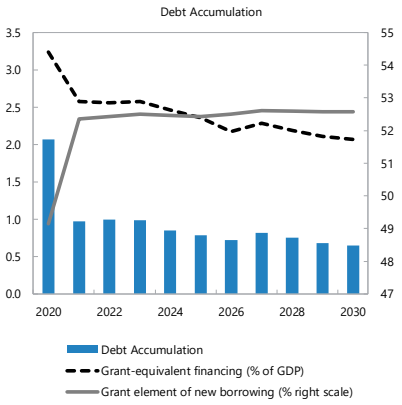
[4] IMF Country Report No 20/96, April 2020

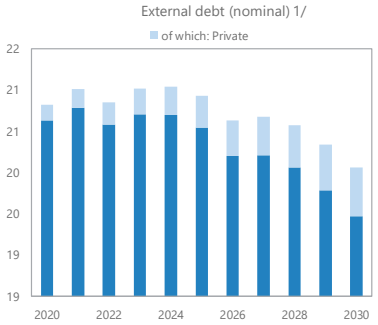
Table 1. Nepal: External Debt Sustainability Framework, Baseline Scenario, 2019-2040

(In percent of GDP, unless otherwise indicated)

	Actual 9/	Projections								Average 8/	
	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
External debt (nominal) 1/	**17.1**	**20.8**	**21.0**	**20.8**	**21.0**	**21.0**	**20.9**	**20.1**	**14.9**	**18.7**	**20.7**
of which: public and publicly guaranteed (PPG)	**17.0**	**20.6**	**20.8**	**20.6**	**20.7**	**20.7**	**20.5**	**19.5**	**13.9**	**18.7**	**20.3**
Change in external debt	-0.2	3.7	0.2	-0.2	0.2	0.0	-0.1	-0.3	-0.8		
Identified net debt-creating flows	**6.4**	**7.1**	**5.9**	**4.0**	**3.6**	**3.2**	**2.6**	**2.0**	**2.7**	**-2.6**	**3.3**
Non-interest current account deficit	**7.6**	**7.4**	**6.9**	**5.6**	**5.0**	**4.6**	**4.1**	**3.5**	**4.1**	**-0.6**	**4.7**
Deficit in balance of goods and services	37.5	31.0	31.1	29.1	28.2	27.4	26.6	23.8	15.9	**29.8**	**27.0**
Exports	8.7	7.7	7.7	8.3	8.5	8.6	8.7	9.1	10.1		
Imports	46.2	38.7	38.8	37.5	36.6	36.0	35.3	32.9	26.0		
Net current transfers (negative = inflow)	-28.7	-22.3	-22.8	-22.2	-21.8	-21.5	-21.1	-18.9	-15.8	**-29.2**	**-20.9**
of which: official	-1.0	-1.1	-1.2	-1.2	-1.2	-1.2	-1.2	-1.1	-1.1		
Other current account flows (negative = net inflow)	-1.3	-1.3	-1.3	-1.3	-1.4	-1.4	-1.4	-1.4	4.0	**-1.2**	**-1.4**
Net FDI (negative = inflow)	**-0.4**	**-0.3**	**-0.5**	**-0.5**	**-0.5**	**-0.6**	**-0.7**	**-0.8**	**-0.8**	**-0.4**	**-0.6**
Endogenous debt dynamics 2/	**-0.8**	**0.0**	**-0.5**	**-1.1**	**-0.9**	**-0.8**	**-0.8**	**-0.8**	**-0.6**		
Contribution from nominal interest rate	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2		
Contribution from real GDP growth	-1.2	-0.2	-0.7	-1.2	-1.0	-1.0	-1.0	-1.0	-0.8		
Contribution from price and exchange rate changes	0.2	...	...	...	...	...	...	...	...		
Residual 3/	**-6.6**	**-3.4**	**-5.7**	**-4.2**	**-3.5**	**-3.1**	**-2.7**	**-2.2**	**-3.6**	**1.4**	**-3.1**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**12.4**	**13.8**	**13.7**	**13.5**	**13.4**	**13.3**	**13.1**	**11.9**	**8.7**		
PV of PPG external debt-to-exports ratio	**142.7**	**179.2**	**177.9**	**161.3**	**158.1**	**154.6**	**150.0**	**131.0**	**85.9**		
PPG debt service-to-exports ratio	**8.0**	**9.4**	**9.4**	**8.6**	**8.7**	**9.0**	**8.5**	**8.2**	**5.3**		
PPG debt service-to-revenue ratio	**2.8**	**3.2**	**3.1**	**3.0**	**3.1**	**3.2**	**3.1**	**3.1**	**2.2**		
Gross external financing need (Million of U.S. dollars)	2427.8	2570.2	2532.3	2264.6	2228.0	2215.0	2111.7	2745.7	6809.1		
Key macroeconomic assumptions											
Real GDP growth (in percent)	7.1	1.0	3.5	6.5	5.4	5.2	5.2	5.2	5.2	**4.9**	**4.8**
GDP deflator in US dollar terms (change in percent)	-1.3	3.5	3.8	2.8	2.4	2.0	2.0	2.3	2.3	**4.3**	**2.6**
Effective interest rate (percent) 4/	0.8	0.9	0.9	0.9	1.0	1.0	1.0	1.0	1.2	**1.0**	**1.0**
Growth of exports of G&S (US dollar terms, in percent)	2.5	-7.0	7.8	18.3	9.5	8.8	9.1	8.7	8.8	**5.6**	**8.1**
Growth of imports of G&S (US dollar terms, in percent)	5.6	-12.3	7.6	5.7	5.4	5.5	5.2	6.5	6.7	**12.9**	**4.4**
Grant element of new public sector borrowing (in percent)	...	49.2	52.4	52.4	52.5	52.5	52.4	52.6	52.1	**...**	**52.2**
Government revenues (excluding grants, in percent of GDP)	24.8	22.8	23.7	23.7	23.7	23.8	23.8	23.9	24.1	**19.3**	**23.7**
Aid flows (in Million of US dollars) 5/	368.3	977.1	892.1	970.0	1053.1	1078.9	1108.5	1411.9	2143.1		
Grant-equivalent financing (in percent of GDP) 6/	...	3.2	2.6	2.6	2.6	2.5	2.4	2.1	1.5	**...**	**2.4**
Grant-equivalent financing (in percent of external financing) 6/	...	60.6	72.3	72.0	71.6	72.5	73.4	72.6	84.4	**...**	**70.8**
Nominal GDP (Million of US dollars)	30690.3	32084.0	34460.5	37719.9	40697.9	43685.5	46876.7	68099.5	142125.8		
Nominal dollar GDP growth	5.7	4.5	7.4	9.5	7.9	7.3	7.3	7.6	7.6	**9.4**	**7.5**
Memorandum items:											
PV of external debt 7/	12.5	14.0	14.0	13.7	13.7	13.6	13.5	12.5	9.7		
In percent of exports	144.4	181.6	180.9	164.5	161.7	158.6	154.4	137.5	95.8		
Total external debt service-to-exports ratio	8.0	11.3	11.8	11.2	11.8	12.5	12.5	14.2	14.7		
PV of PPG external debt (in Million of US dollars)	3790.4	4425.3	4737.0	5078.7	5450.7	5795.5	6137.4	8111.3	12314.4		
(PVt-PVt-1)/GDPt-1 (in percent)		2.1	1.0	1.0	1.0	0.8	0.8	0.6	0.2		
Non-interest current account deficit that stabilizes debt ratio	7.8	3.7	6.7	5.7	4.8	4.6	4.2	3.8	4.9		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (ie. changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
The residuals in this table are mainly related to reserve drawdown and net errors and omissions. For example, there were reserve drawdown of 4.2% of GDP and positive net E&Os of 2.7% of GDP in FY2018/19.
4/ Current-year interest payments divided by previous period debt stock.
5/ Defined as grants, concessional loans, and debt relief.
6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).
7/ Assumes that PV of private sector debt is equivalent to its face value.
8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.
9/ Fiscal year. For example, year 2019 means FY2018/19 (mid-July 2018 to mid-July 2019).

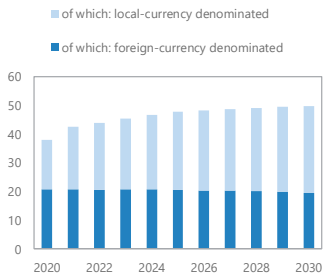
Table 2. Nepal: Public Sector Debt Sustainability Framework, Baseline Scenario, 2019-2040

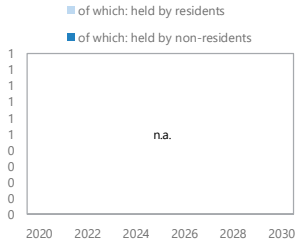
(In percent of GDP, unless otherwise indicated)

	Actual 8/	Projections								Average 6/	
	2019	2020	2021	2022	2023	2024	2025	2030	2040	Historical	Projections
Public sector debt 1/	**30.1**	**38.0**	**42.4**	**43.8**	**45.3**	**46.7**	**47.8**	**49.7**	**50.2**	**30.6**	**46.3**
of which: external debt	**17.0**	**20.6**	**20.8**	**20.6**	**20.7**	**20.7**	**20.5**	**19.5**	**13.9**	**18.7**	**20.3**
							27.3				
Change in public sector debt	-0.2	7.9	4.5	1.4	1.5	1.3	1.1	0.3	-0.2		
Identified debt-creating flows	**0.8**	**5.3**	**3.6**	**0.5**	**0.6**	**0.4**	**0.2**	**-0.5**	**-0.8**	**-2.0**	**0.8**
Primary deficit	**4.0**	**6.6**	**5.8**	**3.8**	**3.5**	**3.2**	**3.1**	**2.6**	**2.4**	**0.5**	**3.6**
Revenue and grants	26.0	24.0	25.2	25.2	25.1	25.2	25.2	25.1	25.3	**21.3**	**25.1**
of which: grants	1.2	1.2	1.5	1.5	1.4	1.4	1.4	1.2	1.2		
Primary (noninterest) expenditure	30.0	30.6	31.0	29.0	28.6	28.4	28.3	27.7	27.7	**21.9**	**28.6**
Automatic debt dynamics	**-3.2**	**-1.3**	**-2.2**	**-3.3**	**-2.9**	**-2.8**	**-2.9**	**-3.1**	**-3.2**		
Contribution from interest rate/growth differential	-2.5	-1.3	-2.2	-3.3	-2.9	-2.8	-2.9	-3.1	-3.2		
of which: contribution from average real interest rate	-0.5	-1.0	-0.9	-0.7	-0.7	-0.6	-0.6	-0.6	-0.7		
of which: contribution from real GDP growth	-2.0	-0.3	-1.3	-2.6	-2.3	-2.3	-2.3	-2.5	-2.5		
Contribution from real exchange rate depreciation	-0.7	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual 7/	**-0.9**	**2.6**	**0.8**	**0.9**	**0.9**	**1.0**	**0.9**	**0.8**	**0.5**	**0.9**	**1.0**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**25.1**	**31.4**	**35.6**	**37.0**	**38.3**	**39.5**	**40.6**	**42.4**	**45.1**		
PV of public debt-to-revenue and grants ratio	**96.4**	**130.8**	**141.5**	**146.7**	**152.4**	**156.8**	**161.1**	**168.8**	**177.9**		
Debt service-to-revenue and grants ratio 3/	**24.4**	**24.3**	**24.2**	**28.5**	**30.6**	**33.5**	**36.8**	**40.6**	**46.8**		
Gross financing need 4/	10.3	12.4	11.9	11.0	11.2	11.7	12.4	12.8	14.2		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	7.1	1.0	3.5	6.5	5.4	5.2	5.2	5.2	5.2	**4.9**	**4.8**
Average nominal interest rate on external debt (in percent)	0.8	0.8	0.9	0.9	0.9	0.9	1.0	0.9	0.9	**1.0**	**0.9**
Average real interest rate on domestic debt (in percent)	-2.6	-6.3	-3.9	-2.6	-2.1	-1.6	-1.5	-1.5	-1.5	**-3.2**	**-2.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	-4.7	...	...	...	...	...	...	...	...	**-2.2**	**...**
Inflation rate (GDP deflator, in percent)	6.8	6.7	6.5	5.8	5.6	5.3	5.3	5.6	5.6	**8.1**	**5.8**
Growth of real primary spending (deflated by GDP deflator, in percent)	2.2	3.1	4.9	-0.3	3.9	4.5	4.8	5.2	5.2	**10.6**	**4.1**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	4.1	-1.3	1.3	2.4	2.0	1.9	2.0	2.3	2.6	**3.5**	**1.8**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus extra budgetary funds, central bank, government-guaranteed debt, non-guaranteed SOE debt . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

7/ The residuals in this table are mainly related to net acquisition of financial assets. For example, the projection of net acquisition of financial assets in FY2019/20 is 1.6% of GDP.

8/ Fiscal year. For example, year 2019 means FY2018/19 (mid-July 2018 to mid-July 2019).

Figure 1. Nepal: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2020-2030

Customization of Default Settings

	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	No	No
Commodity price	n.a.	n.a.
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*

	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	0.9%	0.9%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	36	36
Avg. grace period	6	6

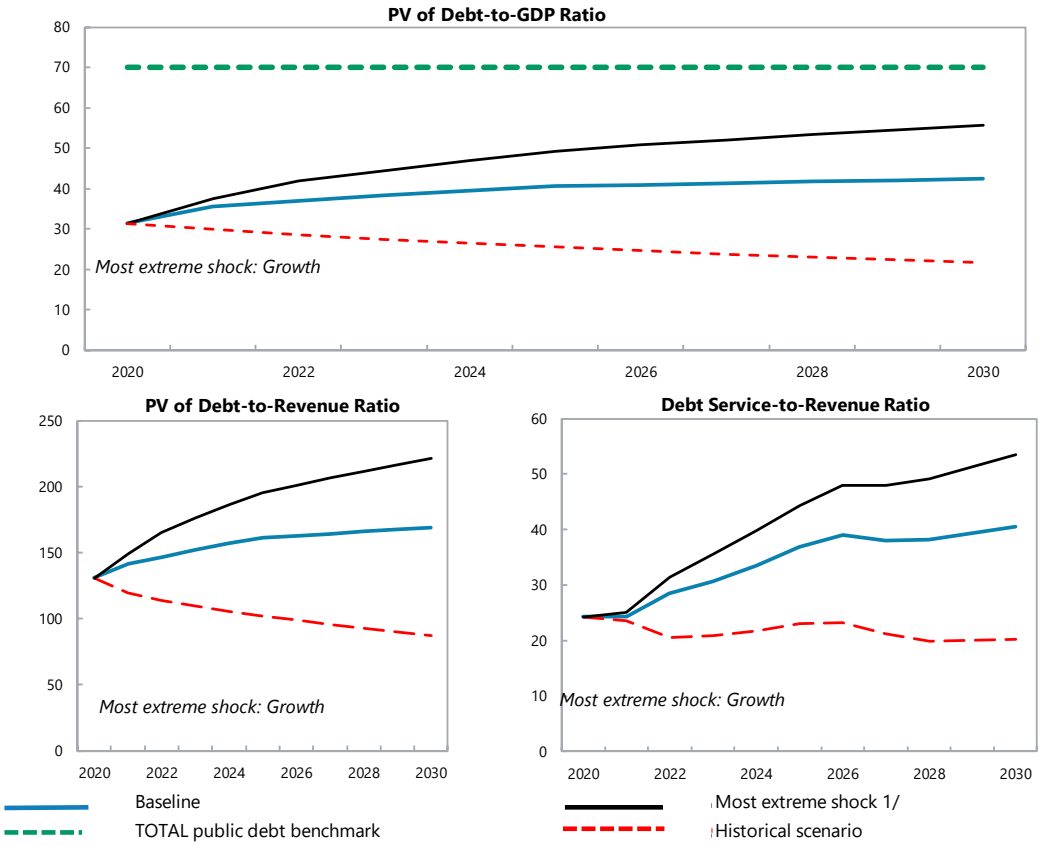
* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

Figure 2. Nepal: Indicators of Public Debt Under Alternative Scenarios, 2020-2030



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	16%	16%
Domestic medium and long-term	57%	57%
Domestic short-term	27%	27%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	0.9%	0.9%
Avg. maturity (incl. grace period)	36	36
Avg. grace period	6	6
Domestic MLT debt		
Avg. real interest rate on new borrowing	-1.8%	-1.8%
Avg. maturity (incl. grace period)	7	7
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	-1.8%	-1.8%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

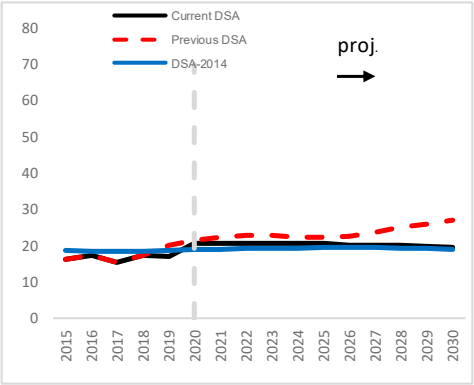
Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2030. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

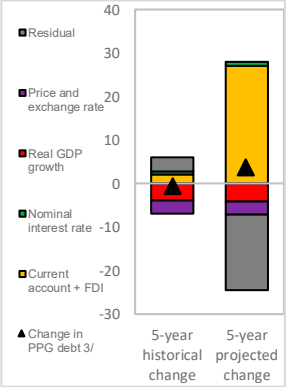
Figure 3. Nepal: Drivers of Debt Dynamics - Baseline Scenario

External debt

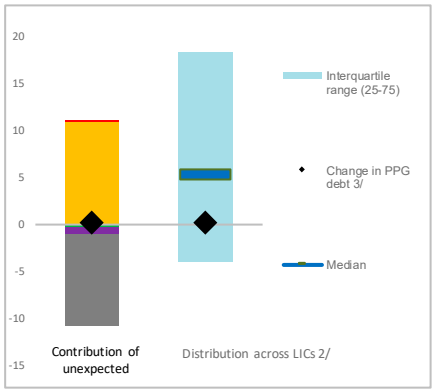
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

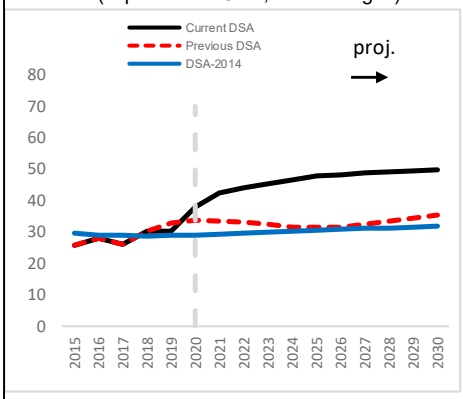


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

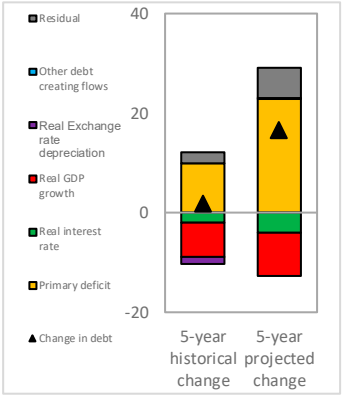


Public debt

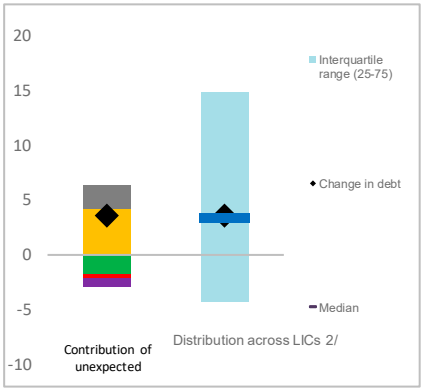
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



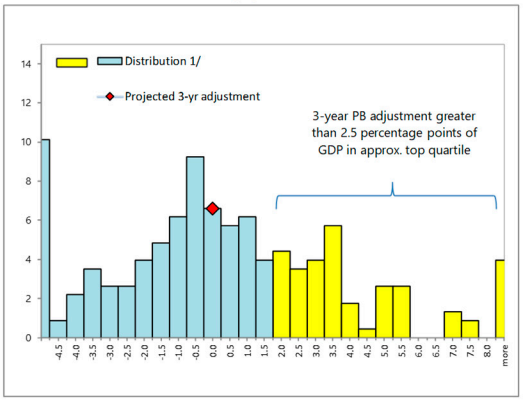
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

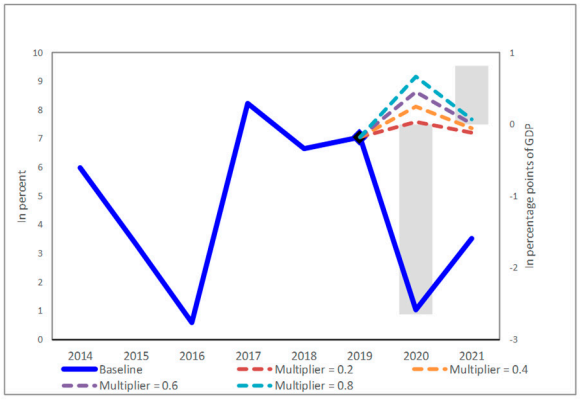
Figure 4. Nepal: Realism tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



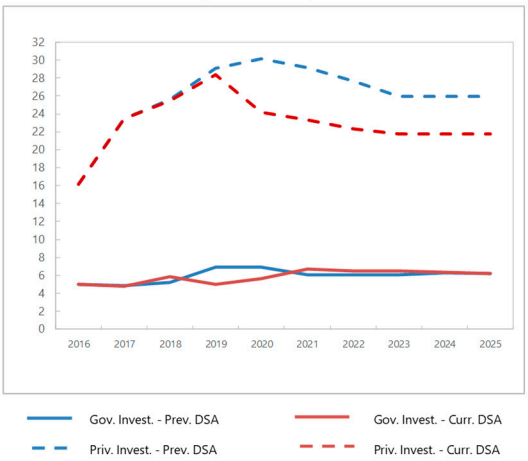
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

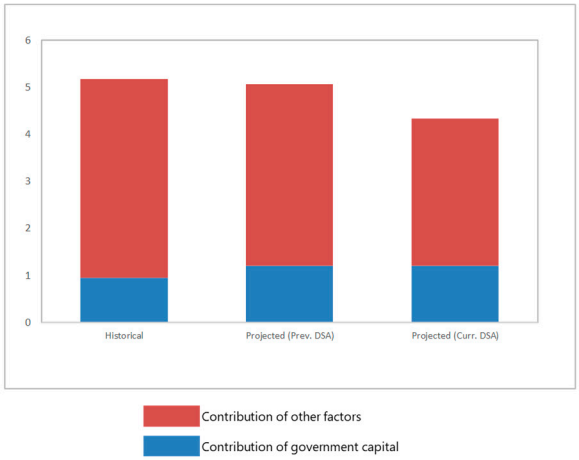


Table 3. Nepal: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2020-2030
(In percent)

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of debt-to GDP ratio											
Baseline	14	14	13	13	13	13	13	13	12	12	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	14	10	7	4	2	0	-2	-3	-5	-6	-7
B. Bound Tests											
B1. Real GDP growth	14	14	14	14	14	14	14	14	13	13	13
B2. Primary balance	14	14	14	14	14	14	14	13	13	13	13
B3. Exports	14	14	15	15	15	14	14	14	14	13	13
B4. Other flows 3/	14	16	17	17	17	17	16	16	16	15	15
B5. Depreciation	14	17	12	12	12	12	12	12	12	12	12
B6. Combination of B1-B5	14	16	17	16	16	16	16	15	15	15	14
C. Tailored Tests											
C1. Combined contingent liabilities	14	14	14	14	14	14	14	14	14	14	13
C2. Natural disaster	14	15	15	15	15	15	15	15	14	14	14
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	179	178	161	158	154	150	146	143	139	135	131
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	179	127	81	50	22	-2	-21	-38	-54	-67	-78
B. Bound Tests											
B1. Real GDP growth	179	178	161	158	154	150	146	143	139	135	131
B2. Primary balance	179	180	167	165	162	158	154	152	149	145	140
B3. Exports	179	202	239	234	228	221	215	210	204	197	190
B4. Other flows 3/	179	204	209	203	198	191	185	181	175	168	161
B5. Depreciation	179	178	116	115	114	111	108	107	105	104	102
B6. Combination of B1-B5	179	214	186	225	219	212	206	201	195	188	181
C. Tailored Tests											
C1. Combined contingent liabilities	179	187	172	170	167	163	160	157	154	151	146
C2. Natural disaster	179	194	180	178	176	172	169	167	165	161	158
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	9	9	8	9	9	8	9	9	9	9	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	9	9	8	7	7	6	6	6	4	2	1
B. Bound Tests											
B1. Real GDP growth	9	9	8	9	9	8	9	9	9	9	8
B2. Primary balance	9	9	9	9	9	9	9	9	9	9	9
B3. Exports	9	10	12	12	12	12	13	13	12	13	12
B4. Other flows 3/	9	9	9	9	10	9	10	10	10	11	10
B5. Depreciation	9	9	8	8	8	8	9	9	8	6	6
B6. Combination of B1-B5	9	10	11	11	11	11	12	12	12	12	11
C. Tailored Tests											
C1. Combined contingent liabilities	9	9	9	9	9	9	10	9	9	9	8
C2. Natural disaster	9	10	9	9	9	9	10	10	9	9	9
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	3	3	3	3	3	3	3	3	3	3	3
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	3	3	3	3	2	2	2	2	1	1	0
B. Bound Tests											
B1. Real GDP growth	3	3	3	3	3	3	4	4	3	3	3
B2. Primary balance	3	3	3	3	3	3	3	3	3	3	3
B3. Exports	3	3	3	3	3	3	3	3	3	4	3
B4. Other flows 3/	3	3	3	3	3	3	4	4	4	4	4
B5. Depreciation	3	4	4	4	4	4	4	4	4	3	3
B6. Combination of B1-B5	3	3	3	3	4	3	4	4	4	4	4
C. Tailored Tests											
C1. Combined contingent liabilities	3	3	3	3	3	3	3	3	3	3	3
C2. Natural disaster	3	3	3	3	3	3	3	3	3	3	3
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Nepal: Sensitivity Analysis for Key Indicators of Public Debt , 2020-2030

	Projections 1/										
	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
PV of Debt-to-GDP Ratio											
Baseline	31	36	37	38	39	41	41	41	42	42	42
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	31	30	29	27	26	26	25	24	23	22	22
B. Bound Tests											
B1. Real GDP growth	31	38	42	44	47	49	51	52	53	55	56
B2. Primary balance	31	38	42	43	44	45	45	45	45	45	45
B3. Exports	31	36	38	40	41	42	42	43	43	43	43
B4. Other flows 3/	31	38	41	42	43	44	44	45	45	45	45
B5. Depreciation	31	37	36	36	37	37	36	36	35	35	34
B6. Combination of B1-B5	31	36	38	38	39	40	40	40	41	41	41
C. Tailored Tests											
C1. Combined contingent liabilities	31	44	44	45	46	46	46	46	46	46	46
C2. Natural disaster	31	46	46	47	48	49	49	49	49	49	50
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio											
Baseline	131	141	147	152	157	161	163	164	166	167	169
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	131	119	114	110	105	102	99	95	92	90	87
B. Bound Tests											
B1. Real GDP growth	131	149	165	176	186	195	201	206	212	217	221
B2. Primary balance	131	152	166	170	173	177	177	177	178	179	179
B3. Exports	131	143	152	158	162	166	168	169	170	172	173
B4. Other flows 3/	131	150	163	168	172	176	177	178	179	179	180
B5. Depreciation	131	147	143	146	146	146	144	142	140	138	137
B6. Combination of B1-B5	131	144	150	151	154	157	159	160	161	163	164
C. Tailored Tests											
C1. Combined contingent liabilities	131	175	176	180	182	184	184	184	184	184	184
C2. Natural disaster	131	181	184	188	191	194	195	195	196	197	197
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	24	24	28	31	33	37	39	38	38	39	41
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2020-2030 2/	24	24	21	21	22	23	23	21	20	20	20
B. Bound Tests											
B1. Real GDP growth	24	25	31	36	40	44	48	48	49	51	53
B2. Primary balance	24	24	32	37	38	41	43	42	42	43	43
B3. Exports	24	24	28	31	34	37	39	38	38	40	41
B4. Other flows 3/	24	24	29	31	34	37	39	38	39	40	41
B5. Depreciation	24	23	27	27	31	34	37	36	36	37	38
B6. Combination of B1-B5	24	24	28	31	33	36	39	38	38	39	40
C. Tailored Tests											
C1. Combined contingent liabilities	24	24	41	38	39	42	45	44	44	44	44
C2. Natural disaster	24	25	43	40	41	45	47	46	47	47	47
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.